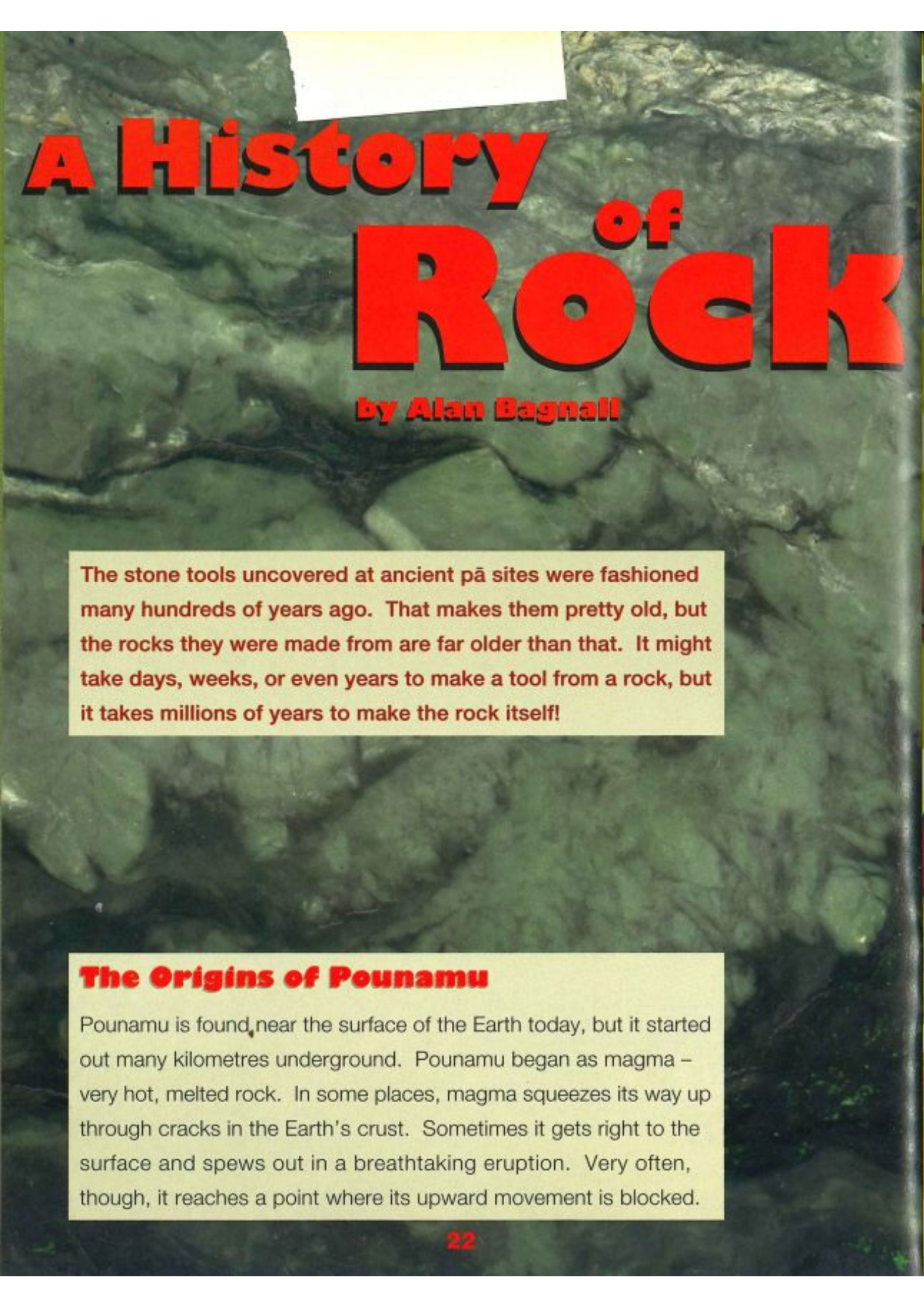




A History of Rock

by Alan Bagnall

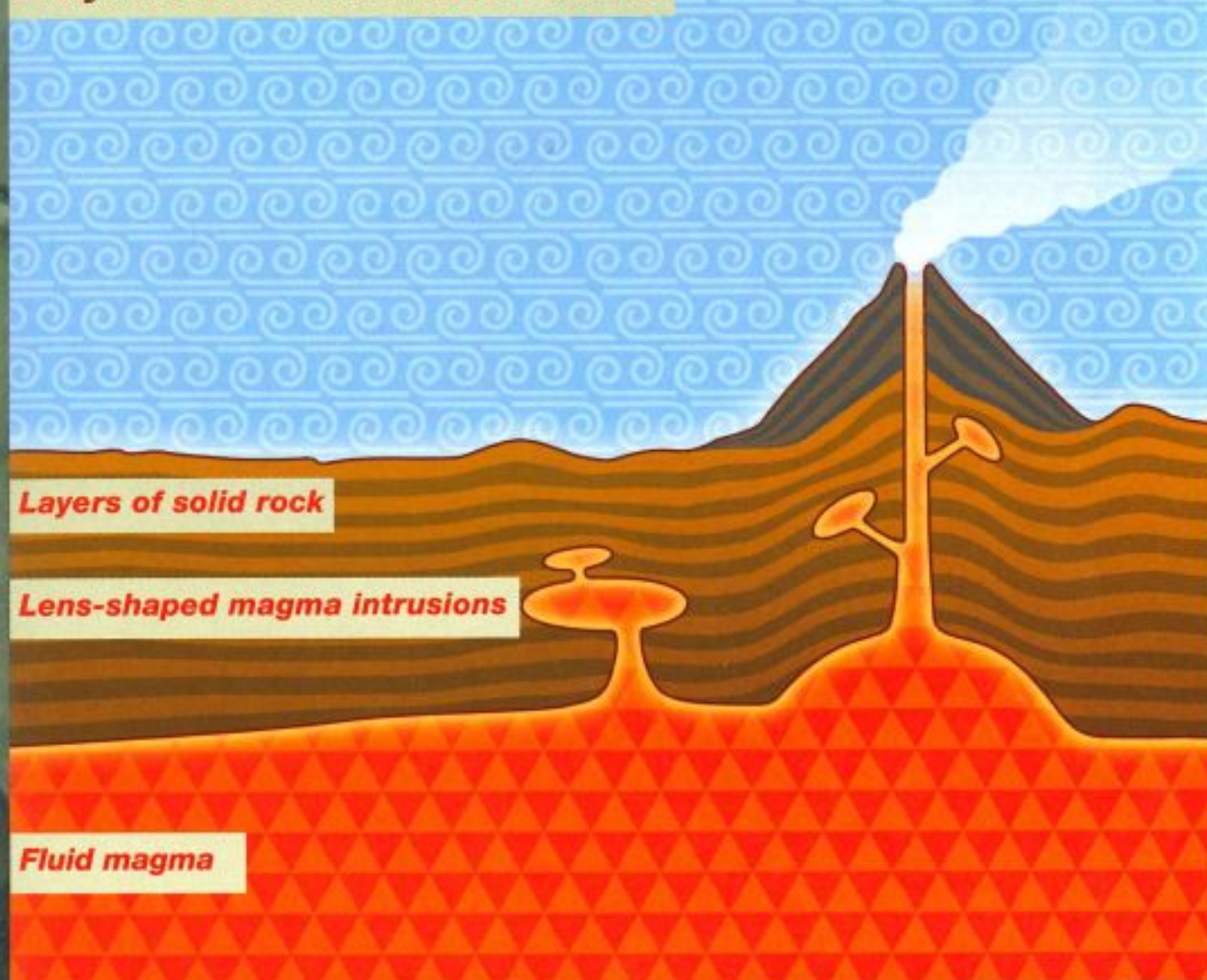
The stone tools uncovered at ancient pā sites were fashioned many hundreds of years ago. That makes them pretty old, but the rocks they were made from are far older than that. It might take days, weeks, or even years to make a tool from a rock, but it takes millions of years to make the rock itself!

The Origins of Pounamu

Pounamu is found near the surface of the Earth today, but it started out many kilometres underground. Pounamu began as magma – very hot, melted rock. In some places, magma squeezes its way up through cracks in the Earth's crust. Sometimes it gets right to the surface and spews out in a breathtaking eruption. Very often, though, it reaches a point where its upward movement is blocked.

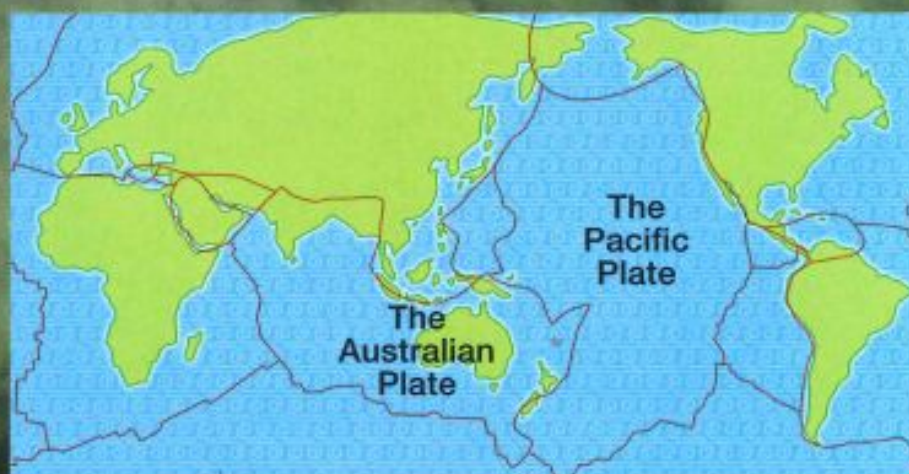
If this happens, pressure can build up so much that the magma spreads out horizontally, forcing its way between layers of rock. Then it slowly cools and becomes a deposit of solid rock called an "intrusion". (Such deposits are often called "lenses" because they're often shaped like a lens that's curved on both sides.)

A stylised cross-section of a volcano



Usually a lens of rock remains pretty much unchanged. However, in some underground areas, the heat and pressure are extreme – so extreme that, over millions of years, the rock gradually changes. It turns from a dull, uninteresting material into the beautiful green treasure we know as pounamu. This process occurred many millions of years ago – during the Cretaceous Period when dinosaurs flourished.

Pounamu is found only in the Southern Alps. Here the Earth's surface is slowly buckling upwards to form the mountainous spine of the island. If we could look below the surface, we'd see what's happening. The Earth seems solid when you're standing on it. However, at a depth of about 40 to 70 kilometres, the Earth's solid outer crust ends and the mantle begins. The rock of the mantle is incredibly hot and under huge pressure. The pressure keeps the mantle solid, but the heat is enough to make the rock "plastic". In other words, the rock of the mantle moves in the same way that chewy toffee will bend if you press it really hard. So the Earth is a bit like a sweet that has hard, brittle candy on the outside and a layer of slow-moving, chewy toffee below this. To complete the picture, imagine that the candy surface of the sweet is cracked into a number of separate sections. On the Earth, these sections are called tectonic plates.



Tectonic plates

In the South Island, a huge section of the Earth's crust called the Pacific Plate is very slowly sliding up over a section of the Australian Plate. The upward movement of the Pacific Plate is creating the Southern Alps. This mountain building is offset by erosion that wears away at the mountains as they rise. Over the last 6 million years, these mountain-building processes have been bringing pounamu lenses up from the depths and exposing them at the surface.

A stylised cross-section of the Alpine Fault

West

The section of the Pacific Plate that has already eroded away

East

The Australian Plate

The Pacific Plate

Lenses of pounamu coming to the surface

The mantle

Solid rock melting into the mantle

The Origins of Other Rocks

Volcanic glass also starts out deep in the Earth as magma, which eventually makes its way right up to the surface. There, it cools and solidifies very quickly. When magma cools very quickly, it forms a type of glass. If it cooled deeper in the Earth, it would solidify only very gradually. Then a very different type of rock would result.

Sandstone and **mudstone** don't develop from magma. Rather, they start out as sandy mud that builds up, layer upon layer, at the bottom of lakes or seas. Over very long periods of time, this mud hardens and slowly turns into a rock called sandstone. (It's called siltstone or mudstone if it has very fine particles.) Māori used sandstone for making patu (flat clubs for warfare). They also used slabs of it as grinding stones for shaping and polishing other rocks.

The **argillite** that Māori used for adze-making started off as ordinary siltstone or mudstone. Then extreme underground heat and pressure very gradually turned it into a heavier, harder rock.

The Alpine Fault

The Australian Plate



The Pacific Plate

The Three Main Rock Types

When a rock has been totally transformed by extreme physical forces, we say that it has "metamorphosed". (You might have heard people use the same word to describe a caterpillar turning into a butterfly.) Pounamu and argillite are **metamorphic rocks**.

When mud and silt settle at the bottom of a waterway, we call the layer of material a sediment. Such layers of mud may become hardened and compressed into what we call a **sedimentary rock**. Sandstone and mudstone are sedimentary rocks.

Igneous rocks such as basalt result when magma moves up from very deep in the Earth. Rock forms when magma cools and solidifies underground – or above ground if there's a volcanic eruption.

So when you look at a landscape, it may seem still and unchanging. But don't be fooled. Rock, soil, and water are constantly on the move. Today's landscape is like a single frame from a very long movie. If we could play the movie on fast forward, we'd see the Earth's surface buckle and writhe. We could watch it build up and break down as rocks and landscapes are formed, then worn away. We would then recognise the Earth as a giant rock factory in full production.

Pounamu



Argillite



Sandstone



Basalt

